DATABASE FORENSICS

# Weekly task list

A fictional SQLite order ledger includes an audit trail. Examine a preserved copy using read-only access and compare recorded changes with final rows; this is not a production database investigation.

## The investigation question

Are the final ledger rows internally consistent with their keys, arithmetic and audit transitions?

## Your four weekly tasks

## Week 1 | Read-only database intake

Inventory training.db and schema.sql, compute SHA-256 values and preserve originals. Open only a working copy in read-only mode. List tables, column meanings, primary keys and relationships.

Output: schema_map.pdf, evidence_register.csv and read_only_method.pdf

## Week 2 | Integrity and consistency checks

Count rows by table and check duplicate order IDs, null required fields and references to missing customers. Compare recorded order totals to quantities and unit prices, keeping the exact SELECT statements used.

Output: checks.sql, integrity_results.csv and discrepancy_notes.pdf

## Week 3 | Audit trail reconstruction

Join audit records to orders using their identifiers and sort changes by timestamp. Compare before_value, after_value and the final order amount. Flag a missing or inconsistent audit transition.

Output: audit_timeline.csv, reconstruction.sql and limitations.pdf

## Week 4 | Database findings report

Write a 4-6 page report with schema, read-only method, integrity checks and a reconstruction of two selected orders. Include query text, result excerpts, row IDs and the difference between observed inconsistency and attribution.

Output: database_report.pdf, queries.sql and results.zip

Read pages 2-5 for the full checklists. Submission rules follow on page 6; requirements and installation are on pages 7-8. Worked methods and three reading downloads are on page 9. The complete roadmap is the final page.

WEEK 1 | TASK INSTRUCTIONS

# Read-only database intake

## Objective

Create a trustworthy baseline before interpreting any record.

## Do this in order

1. Inventory training.db and schema.sql, compute SHA-256 values and preserve originals.

2. Open only a working copy in read-only mode. List tables, column meanings, primary keys and relationships.

3. Record engine/tool versions and explain the absence of transaction logs, original host context and a historical snapshot.

4. Create a case note naming training.db and schema.sql. For each, record evidence ID, original filename, byte size, SHA-256 and where the PFSA ZIP was downloaded. Keep the supplied manifest with the ZIP.

5. Open a working copy read-only and record that mode. Do not click Write Changes or run an UPDATE, DELETE, VACUUM or repair operation on the evidence.

6. Recheck one result independently: return to the source, repeat the calculation or lookup, and record whether it agrees. Keep a short correction log instead of silently revising previous findings.

## Required hand-in

schema_map.pdf, evidence_register.csv and read_only_method.pdf

Name the upload PFSA_<student-id>_database_W1.pdf or .zip. A ZIP may contain the report plus CSV, SQL, images or other permitted supporting work. Do not rename a non-PDF file to .pdf. Keep the one portal upload below 10 MB.

## Portal description and acceptance check

Write at least 200 genuine words in the description box: what you examined, how you processed it, exact evidence references, results, uncertainty and what you learned. Before submitting, open the uploaded work locally, confirm all filenames resolve, and compare original-file hashes with your intake record.

WEEK 2 | TASK INSTRUCTIONS

# Integrity and consistency checks

## Objective

Turn source records into a documented examination worksheet.

## Do this in order

1. Count rows by table and check duplicate order IDs, null required fields and references to missing customers.

2. Compare recorded order totals to quantities and unit prices, keeping the exact SELECT statements used.

3. Distinguish intentionally unusual training records from conclusions about misconduct; explain limitations of the supplied schema.

4. Keep all original columns from training.db and schema.sql. Add new analysis columns rather than replacing source identifiers or timestamps. Preserve the query, formula or filter used.

5. Preserve exact SELECT statements beside results. If a tool reformats values, record the display format and compare numeric values explicitly.

6. Recheck one result independently: return to the source, repeat the calculation or lookup, and record whether it agrees. Keep a short correction log instead of silently revising previous findings.

## Required hand-in

checks.sql, integrity_results.csv and discrepancy_notes.pdf

Name the upload PFSA_<student-id>_database_W2.pdf or .zip. A ZIP may contain the report plus CSV, SQL, images or other permitted supporting work. Do not rename a non-PDF file to .pdf. Keep the one portal upload below 10 MB.

## Portal description and acceptance check

Write at least 200 genuine words in the description box: what you examined, how you processed it, exact evidence references, results, uncertainty and what you learned. Before submitting, open the uploaded work locally, confirm all filenames resolve, and compare original-file hashes with your intake record.

WEEK 3 | TASK INSTRUCTIONS

# Audit trail reconstruction

## Objective

Correlate events without hiding uncertainty or alternative explanations.

## Do this in order

1. Join audit records to orders using their identifiers and sort changes by timestamp.

2. Compare before_value, after_value and the final order amount. Flag a missing or inconsistent audit transition.

3. Explain why an application-level audit record is not a complete database transaction log or proof of actor identity.

4. Give every derived timeline event an exact source file and row or event ID. Record raw time, UTC time, conversion rule, evidence meaning and uncertainty in separate fields.

5. Use a LEFT JOIN to retain orders whose customer is missing. An INNER JOIN would silently remove those rows from the result.

6. Recheck one result independently: return to the source, repeat the calculation or lookup, and record whether it agrees. Keep a short correction log instead of silently revising previous findings.

## Required hand-in

audit_timeline.csv, reconstruction.sql and limitations.pdf

Name the upload PFSA_<student-id>_database_W3.pdf or .zip. A ZIP may contain the report plus CSV, SQL, images or other permitted supporting work. Do not rename a non-PDF file to .pdf. Keep the one portal upload below 10 MB.

## Portal description and acceptance check

Write at least 200 genuine words in the description box: what you examined, how you processed it, exact evidence references, results, uncertainty and what you learned. Before submitting, open the uploaded work locally, confirm all filenames resolve, and compare original-file hashes with your intake record.

WEEK 4 | TASK INSTRUCTIONS

# Database findings report

## Objective

Deliver a clear, reproducible report that answers the case question.

## Do this in order

1. Write a 4-6 page report with schema, read-only method, integrity checks and a reconstruction of two selected orders.

2. Include query text, result excerpts, row IDs and the difference between observed inconsistency and attribution.

3. Re-hash the preserved original and provide a reproducibility appendix for all SELECT queries.

4. Write an executive summary first, then scope, evidence register, tool versions, reproducible methods, findings, alternatives, limitations and conclusion. Put detailed tables and queries in appendices.

5. Link audit.order_id to orders.id and compare transition values in time order. An application audit trail is not a full transaction log.

6. Recheck one result independently: return to the source, repeat the calculation or lookup, and record whether it agrees. Keep a short correction log instead of silently revising previous findings.

## Required hand-in

database_report.pdf, queries.sql and results.zip

Name the upload PFSA_<student-id>_database_W4.pdf or .zip. A ZIP may contain the report plus CSV, SQL, images or other permitted supporting work. Do not rename a non-PDF file to .pdf. Keep the one portal upload below 10 MB.

## Portal description and acceptance check

Write at least 200 genuine words in the description box: what you examined, how you processed it, exact evidence references, results, uncertainty and what you learned. Before submitting, open the uploaded work locally, confirm all filenames resolve, and compare original-file hashes with your intake record.

AFTER THE WEEKLY TASK LIST

# Submission & assessment

## What to submit each week

Submit one main project file and a separate portal narrative of at least 200 words. Use a PDF for a readable report; use a ZIP when the report needs supporting CSV, SQL, JSON, screenshots or source code. The portal supports multiple work formats, but it is not a substitute for organising your evidence.

| Item | Required content |
|---|---|
| Main report | Student ID, domain, week, case question, scope, method, findings and limitations. |
| Appendices | Source references, derived tables, saved formulas/queries, tool versions and relevant screenshots. |
| Integrity record | Original file hashes and verification result; explain whether any working file changed. |
| Portal narrative | At least 200 words written in your own words; a report attachment does not replace this field. |

## A useful 200-word narrative structure

Use approximately 40-60 words for scope and inputs, 60-90 for your actual method, 60-90 for supported findings, and 40-60 for uncertainty and learning. These are writing prompts, not text to copy. Mention at least two precise record references and one limit of the evidence. Do not repeat a sentence or paste filler just to reach the counter.

## Marking rubric | 100 marks per week

| Criterion | Marks | What a reviewer looks for |
|---|---|---|
| Integrity and scope | 20 | Correct baseline, preserved originals and a defined question. |
| Method | 25 | Reproducible steps, versions, calculations and saved filters. |
| Analysis | 25 | Supported reasoning and alternatives, with appropriate uncertainty. |
| Evidence references | 20 | Exact file and row/event references for each significant finding. |
| Communication | 10 | Clear report, organised package and meaningful portal description. |

Calendar deadlines and any resubmission instructions are published in the portal by staff. Four submissions unlock the next document stage; payment confirmation, project assessment and document publication remain staff-controlled. Ask Support about an incorrect upload or locked record.

LAPTOP, SOFTWARE & FILES

# Requirements before you begin

## Laptop and workspace

Use a laptop or desktop with a supported Windows, macOS or Linux installation, a keyboard and permission to install the required applications. For these small training exports, 4 GB RAM is a practical minimum; 8 GB is recommended. Reserve at least 2 GB free for tools and working files, with additional space for downloaded books. A phone can read PDFs and check portal status, but a laptop is strongly recommended for evidence tables, file packaging and reproducible work.

## Required software for this domain

DB Browser for SQLite with Open Database Read Only; LibreOffice Calc and Writer; plain-text editor. Python 3 sqlite3 is an optional alternative. PostgreSQL reading is background only; do not migrate this SQLite evidence file into a new database for the core task.

## Common tools and their purpose

| Tool | Use in the assignment |
|---|---|
| LibreOffice Calc | Import CSVs, retain text identifiers, calculate summaries and save derived tables. |
| LibreOffice Writer | Write the case report and export it as a readable PDF. |
| Plain-text editor / VS Code | Inspect JSON, JSONL, SQL and notes without altering formatting in a word processor. |
| Native SHA-256 utility | Create and compare an integrity baseline for each source file. |
| Browser and PDF reader | Read the guide, open references and submit work through the student portal. |

## Training files and folder structure

Download database-training-data.zip from the matching domain card. Its central source files are training.db and schema.sql. Save the original ZIP, extract it once, and create four folders: originals, working, analysis and submission. Put the original extracted files in originals, analysis copies in working, derived outputs in analysis, and only final hand-in material in submission.

## Before the first task

Record your operating system and the installed application versions. Confirm that the file extensions are visible and that you can create a CSV, open a PDF and make a ZIP. Avoid syncing personal or confidential data into this case folder. The training pack is fictional; no purchase, live investigation account or collection from another person is needed.

PRACTICAL SETUP | WINDOWS, MACOS & LINUX

# Installation & first-run checks

## 1 | Install a document and spreadsheet suite

Open the official LibreOffice download page and choose your operating system and CPU architecture. On Windows, download the stable main installer, run it and follow the standard setup prompts; open both Calc and Writer after installation. On macOS, choose Intel or Apple Silicon as appropriate, open the DMG and move LibreOffice to Applications. On Linux, prefer the distribution's supported software manager. Record the version in your case notes.

LibreOffice download and installation instructions

## 2 | Set up a plain-text editor

Download VS Code from its official site if you do not already have a suitable editor. On Windows, the user installer normally avoids a machine-wide installation. Open only your working case folder using File > Open Folder. Inspect JSON or SQL as text, keep automatic file formatting off for evidence copies, and do not add extensions that are not needed. A built-in text editor is sufficient when installing VS Code is not possible.

Official VS Code Windows setup

## 3 | Verify hashes without an extra package

On Windows, open PowerShell in your working case folder and run Get-FileHash -Algorithm SHA256 .\originals\<filename>. On macOS use shasum -a 256 originals/<filename>. On Linux use sha256sum originals/<filename>. Replace the placeholder with one actual filename. Save each digest with its filename, size, date and tool. Repeating the command against the same preserved file should produce the same digest.

## 4 | Domain-specific installation and first run

Download the stable DB Browser for SQLite build for your operating system from the official site. Install or extract it, launch it and use Open Database Read Only to select working/training.db. Check the title/status for read-only mode before examining tables. If that mode is unavailable, use a new working copy and the documented Python alternative with URI mode=ro; do not use a production database.

DB Browser official downloads

Installation checks: create a test worksheet, export one PDF, read one source file from a working copy and reproduce a SHA-256 digest. Save a short tools.txt with versions and any differences from these instructions. Official setup pages are the source of truth if a vendor changes a screen label.

DOMAIN-SPECIFIC GUIDANCE

# Worked method & reading library

## First reproducible examination

Open the working database with the read-only command. In Execute SQL, run SELECT name FROM sqlite_master WHERE type='table'; then SELECT COUNT(*) FROM orders;. Save query text to checks.sql and results to CSV. For the discrepancy check, use SELECT id, quantity*unit_price AS expected_total, total FROM orders WHERE total != quantity*unit_price;. Include the schema and note how currency precision would be handled in a real ledger.

## A limit you must state

A clean integrity check does not prove the business data are correct. Do not treat the supplied audit actor as a verified person or infer deleted rows without suitable evidence.

## Three books & substantial reference downloads

Use the original publisher or author links below. Reference guides and manuals are labelled separately from textbooks. Older editions provide background; they are not a substitute for current tool documentation. Third-party authors retain their rights and do not endorse PFSA.

### 1 | Database Design - 2nd Edition

Adrienne Watt and contributors / BCcampus | Second edition | Open textbook

Study relational design, keys, integrity and SQL concepts.

Open authorised PDF / author download page

### 2 | PostgreSQL 18 Documentation

PostgreSQL Global Development Group | Version 18 manual | Database reference manual

Read the SQL tutorial and transaction concepts; PostgreSQL syntax is not automatically valid in the SQLite exercise.

Open authorised PDF / author download page

### 3 | Foundations of Databases

Serge Abiteboul, Richard Hull and Victor Vianu | Author-provided book draft, 1995 | Author download page / textbook

Use the full PDF link on the authors page for personal study of relational models, keys and dependencies. The authors permit a personal copy but prohibit redistribution.

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Download links checked 13 November 2026. Use the source page if a PDF host changes or restricts automated downloads. Foundations of Databases is linked to the authors download page because its personal-use terms prohibit redistribution. These reading files are not required attachments to your weekly submission.

FINAL PAGE | PLAN, PRACTISE, VERIFY

# Four-week learning roadmap

## Your four-stage learning path

Read-only intake -> relational/arithmetic checks -> audit reconstruction -> database findings report

Use this roadmap to plan the sequence, not to calculate calendar deadlines. Staff publish the actual dates in your portal. Early Start changes when access opens; it does not remove any weekly deliverable or assessment requirement.

| Stage | Suggested focus | Checkpoint before moving on |
|---|---|---|
| Week 1 \| Preserve | Days 1-2: scope and inventory. Days 3-4: integrity and working copy. Days 5-7: review and submission. | Every source has an ID, hash and documented limitation. |
| Week 2 \| Examine | Days 8-9: field definitions. Days 10-11: structured checks. Days 12-14: independent recheck and submission. | Derived tables retain original IDs and reproducible calculations. |
| Week 3 \| Correlate | Days 15-16: time conversion. Days 17-18: source correlation. Days 19-21: alternatives and submission. | Timeline retains raw time and explains ambiguity, not just a neat order. |
| Week 4 \| Communicate | Days 22-23: findings. Days 24-25: report and appendices. Days 26-28: independent review and final hand-in. | A reviewer can repeat the method and trace every significant claim. |

## Daily working habit

Start by reading your last case note and checking the source hash. Work on one defined question, save the method and result, then write a brief end-of-session note: completed work, decisions, outstanding questions and next step. Keep the final report consistent with the latest version of the analysis tables.

## Final readiness gate

Confirm four genuine weekly submissions, complete evidence references, original-file integrity checks and an organised final report. Read staff feedback and resolve requested corrections. Keep a personal copy of the submitted package and the portal acknowledgement. Document eligibility and recommendation remain subject to assessment and the academy's published service terms.

## What you should be able to demonstrate

Explain the supplied evidence boundary, reproduce one key finding, compare a competing explanation and describe which additional evidence would improve confidence. That explanation is more valuable than a screenshot of a tool running.

Student portal and Support